

DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

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ABSTRACT

Parametric imitation learning via behavior cloning remains vulnerable to out-of-distribution states because compounding deployment errors expose the policy to regions unconstrained by expert supervision. We show that this weakness can be directly mitigated by reusing the training data at inference time through a semi-parametric retrieval-based imitation learning policy. We introduce **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, a retrieval-based approach that decisively reparameterizes imitation learning around local neighborhood structure rather than direct global state-to-action mappings. Instead of compressing demonstrations into a single global policy, DARP predicts actions from k -nearest expert neighbors, their actions, and relative distance vectors between neighbor states and query states. DARP preserves the assumptions of standard behavior cloning: it requires no additional data collection, online expert feedback, or task-specific knowledge. Across continuous-control and robotic-manipulation domains, and across low-dimensional and high-dimensional visual representations, DARP delivers consistent 15-46% performance improvements over standard behavior cloning. Code and demos will be released upon acceptance.

1 INTRODUCTION

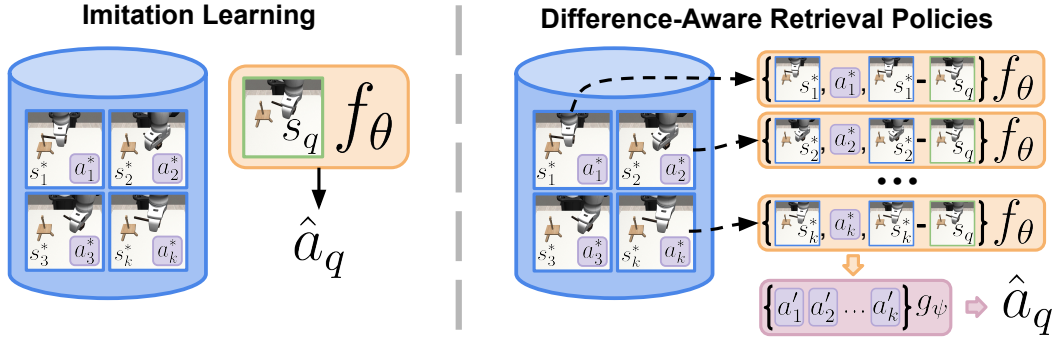


Figure 1: Overview of DARP: Unlike standard BC (left), DARP (right) makes the expert dataset an active inference-time component through a retrieval-based reparameterization centered on difference vectors between query states and retrieved neighbors. Standard behavior cloning uses expert state-action pairs only during training and discards them at inference time, whereas DARP retrieves a local neighborhood of expert state-action pairs around each query point s_q before predicting the action.

Imitation learning via behavior cloning (BC) (Pomerleau, 1991) has become a central recipe for training robots to execute complex, dexterous behaviors from expert demonstrations (Zhao et al., 2023; Chi et al., 2024; Black et al., 2024; Chung et al., 2014). Its simplicity, however, masks a decisive failure mode: BC is brittle under long-horizon closed-loop deployment (Ross et al., 2011). The core issue is *covariate shift*: small action errors compound during rollouts and drive the agent into states that are poorly represented in the demonstration data (Spencer et al., 2021; Ross et al., 2011). In these out-of-distribution regions, standard BC policies can become unstable, producing high-variance behavior that often causes task failure.

The compounding-error problem is well understood, and a large body of work attacks it through additional supervision or stronger interaction assumptions (Ross et al., 2011; Venkatraman et al., 2015; Ke et al., 2024b; Levine et al., 2020). These solutions are powerful but typically leave the standard BC setting by requiring simulators, interactive experts, large quantities of sub-optimal data, or task-specific structure. We instead keep the setting deliberately strict: the learner receives only expert state-action pairs and no additional feedback. This paper answers the resulting question affirmatively: *BC policy variance can be reduced using only the original demonstration dataset.*

Statistically, BC minimizes supervised risk only on expert states. This controls bias on the training distribution but leaves estimator variance unconstrained: in low-density regions of the state space, which closed-loop rollouts commonly encounter, the learned policy can oscillate sharply. Smoothness is the right structural remedy because nearby states should induce compatible actions (Kobayashi, 2022; Asadi et al., 2018; Ke et al., 2024a; Chen et al., 2024). Enforcing that structure suppresses spurious fluctuations and improves rollout stability. Section 4 discusses prior approaches that impose smoothness explicitly.

These approaches can help, but each pays a clear price: augmentation does not guarantee local consistency, global priors can blur distinct behaviors, temporal penalties act along trajectories rather than across the state space, and explicit graph regularizers introduce additional smoothness hyperparameters that must be tuned.

A complementary perspective separates global and local learning. “Global” supervised models (Black et al., 2024; Zhao et al., 2023; Chi et al., 2024) compress the entire demonstration set into one parametric function, a design that is exposed under distribution shift. “Local” methods (Pari et al., 2022; Mansimov & Cho, 2018; Salzberg & Aha, 1994) instead keep the dataset available, consult neighborhoods of similar states, and generate predictions through non-parametric or semi-parametric operations on expert behavior. Locality delivers robustness by grounding predictions in nearby demonstrations, but standard local methods remain limited: performance depends on the distance metric, naive neighborhood averaging can blur distinct actions and mishandle multimodality, and representing neighbors only by absolute states restricts generalization.

We introduce **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, which unifies the robustness of local retrieval with the stability of regularized policy learning. At inference time, DARP (Fig. 1) does not rely solely on a feedforward prediction from the query state. It retrieves k neighbors from the training corpus, predicts candidate actions conditioned on tuples of (neighbor state, associated action, and difference from the query state), and aggregates those candidates in a permutation-invariant manner to produce a robust action. This design grounds predictions in observed data while implicitly enforcing local consistency through parametric action prediction conditioned on retrieved neighbors. We show that this reparameterization reduces variance without any additional assumptions beyond standard Behavior Cloning: no additional data, online supervision, or task-specific knowledge is required. Spectrally, neighbor aggregation approximates a Laplacian filter on the k -NN (k -nearest neighbor) graph of expert states, yielding parameter-free smoothing that adapts to local dataset density and geometry.

We provide theoretical and empirical evidence that DARP substantially improves imitation learning while preserving the assumptions of behavior cloning. Our analysis formalizes the connection to Laplacian regularization and shows that DARP implicitly applies a fixed low-pass spectral filter that suppresses high-frequency variance. Empirically, DARP achieves 15–46% gains over standard behavior cloning across continuous control (MuJoCo), robotic manipulation (Robosuite, Robocasa), and high-dimensional visual imitation tasks (Robosuite with image state). These results establish DARP as a general and scalable architecture that extends naturally to image-based domains and richer policy classes such as transformers and Gaussian mixture models. Our ablations identify the representation and aggregation choices that drive the gains, yielding concrete design principles for retrieval-based sequential decision making.

2 DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

We instantiate a class of imitation learning methods that deliberately combines the strengths of “global” parametric learning and “local” retrieval-based learning. The proposed architecture keeps the training requirements of standard behavior cloning while delivering clear theoretical and empirical

improvements. This section derives the guarantees from first principles. Readers primarily interested in the practical algorithm may skip to Section 2.5 for a self-contained summary of DARP before proceeding to the empirical evaluation in Section 3.

We first define the imitation learning setting (Section 2.1) and introduce a regularized variant of behavior cloning (Section 2.2) that uses dataset geometry to improve variance, generalization, and stability. Section 2.3 then shows that the same benefits can be obtained by changing the policy *architecture* rather than the loss. Finally, Section 2.4 presents DARP, a practical semi-parametric retrieval-augmented architecture that realizes these benefits with modern neural networks and generative policy models.

2.1 PRELIMINARIES: BEHAVIOR CLONING FOR IMITATION LEARNING

We work in the standard imitation learning setting, formalized as a finite-horizon Markov Decision Process (MDP), $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, P_0\}$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, and P_0 is the initial state distribution. A *policy* maps a state to a distribution over actions, $f_\theta : \mathcal{S} \rightarrow \Delta_{\mathcal{A}}$, in service of task-relevant objectives; for brevity, we drop the subscript θ when referring to the function class f . The learner receives expert human-provided demonstrations $\mathcal{D}^*$ as state-action pairs, $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$, with s^* and a^* reserved for expert states and actions. Behavior cloning (Pomerleau, 1991) learns f_θ by casting imitation as supervised learning, $\arg \max_{\theta} \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\log(f_\theta(a^* | s^*))]$. Although f_θ can be any expressive generative policy class (Lipman et al., 2023; Chi et al., 2024), we begin with a Gaussian parameterization for clarity¹.

2.2 WARM-UP: NEIGHBOR MANIFOLD REGULARIZED IMITATION LEARNING

Behavior cloning minimizes supervised imitation loss only on expert states drawn from $\mathcal{D}^*$. That objective does not control policy behavior away from the expert-state manifold. During deployment, accumulated errors can therefore move the agent into out-of-distribution regions where an overparameterized BC policy may act arbitrarily (Ross & Bagnell, 2010).

The missing structure is neighborhood geometry. BC constrains function values at training states but ignores how nearby states and their actions relate on the data manifold. We therefore introduce an explicit regularizer that enforces *local consistency*: nearby expert states should induce compatible actions. This yields the neighborhood-regularized loss ($\mathcal{L}_{\text{MRIL}}$), which augments the standard imitation objective ($\mathcal{L}_{\text{BC}}$) with a smoothness penalty ($\mathcal{L}_S$) that forces predictions to respect dataset geometry rather than relying only on pointwise supervision.

$$\mathcal{L}_{\text{MRIL}}(f) = \underbrace{\mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)]}_{\text{supervised risk}(\mathcal{L}_{\text{BC}})} + \lambda \underbrace{\mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right]}_{\text{smoothness regularizer}(\mathcal{L}_S)}, \quad (1)$$

Here $\ell(f(s^*), a^*)$ is the supervised imitation loss, $\mathcal{N}_k(s^*)$ denotes the k -nearest expert neighbors of s^* , and $w_i(s^*)$ are normalized kernel weights based on state differences, $w_i(s^*) \propto K_\Delta\left(\frac{\|s_i^* - s^*\|}{h}\right)$. As summarized below and proved in Appendix A.1.1, this objective is a concrete form of *manifold regularization*, or Laplacian smoothing: it penalizes high-frequency policy variation across neighborhoods of expert states. This regularization yields provable gains in variance, stability, and generalization.

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ be the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$. Let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider two estimators trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

¹We show that this can be relaxed in Section 2.4

2. MRIL: a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Then, under the smoothness assumption on f , the following hold:

- (i) Variance reduction: The Laplacian penalty in MRIL acts as a data-dependent Tikhonov regularizer, yielding smaller estimator variance than vanilla BC.
- (ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants with arbitrarily large Lipschitz constants between training states.
- (iii) Policy stability: In a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^2$$

accumulates error linearly for vanilla BC, but sublinearly for MRIL, since the smoothness regularizer enforces $\|f(s^*) - f(s'^*)\| = O(\|s^* - s'^*\|)$ for neighbors s^*, s'^* .

Thus MRIL provides stronger generalization and stability guarantees than vanilla BC.

Proof sketch. Appendix A.1.1 gives the full proof; here we give the core argument. The smoothness term is exactly a graph Laplacian penalty on the k -nearest neighbor (k -NN) affinity graph induced by the kernel weights w_i . As the sample size grows, this graph Laplacian penalty converges to the weighted Dirichlet energy (Belkin & Niyogi, 2008; Zhou et al., 2003). Minimizing that energy makes f locally Lipschitz almost everywhere, which delivers smoothness and rollout stability, and simultaneously acts as Tikhonov regularization, reducing estimator variance while keeping bias controlled. $\square$

Intuitively, the smoothness regularizer does more than penalize pairwise neighbor disagreement: it forces the learned policy to respect the underlying data manifold. In high-density directions of the state distribution $p(s)$, it shrinks the local Lipschitz constant of f , so small state changes produce controlled action changes. The resulting policy generalizes more reliably on in-distribution (ID) states and extrapolates more systematically to nearby out-of-distribution (OOD) states.

2.3 IMPLICIT MANIFOLD REGULARIZATION VIA IN-CONTEXT ARCHITECTURES

The MRIL objective demonstrates the value of amortized local learning, but it also exposes two practical costs. First, it introduces a hyperparameter λ that must balance supervised accuracy and smoothness. Second, optimizing a modified regularized objective can alter the optimization landscape relative to standard BC. This motivates a sharper architectural question: *can the benefits of MRIL (Eq. 1) be obtained by changing the policy architecture rather than the objective?*

We answer this question with a retrieval-based architectural change that induces *implicit* manifold regularization (iMRIL) while retaining the standard imitation objective. iMRIL obtains the Laplacian-smoothing benefits of MRIL through the architecture itself, as illustrated in Fig. 2, and removes λ from the training problem. Section 2.4 then turns this principle into the practical DARP algorithm.

iMRIL architecture: The key move in iMRIL is to move neighborhood aggregation from the objective in Eq. 1 into the architecture. Instead of training a standard feedforward predictor $f(s)$ with an explicit smoothness penalty (Eq. 1), we embed neighborhood aggregation directly in the action predictor $\hat{f}$ while keeping the loss equal

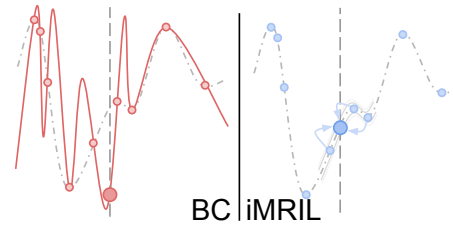


Figure 2: **iMRIL builds Laplacian smoothing directly into the policy architecture, reducing variance and enforcing local consistency**, whereas standard BC lacks a smoothness constraint and can admit arbitrarily jagged function approximations.

² L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

to standard imitation learning. iMRIL learns a per-state predictor f_θ such that the aggregated predictor $\hat{f}(s^*) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*)$ over nearest expert neighbors $\{s_i^*\}_{i \in \mathcal{N}_k(s^*)}$ accurately predicts the expert action a^* . With this parameterization, iMRIL optimizes at training time:

$$\arg \min_{\theta} \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} \left[\left\| \underbrace{\left(\frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*) \right)}_{\hat{f}(s^*)} - a^* \right\|_2 \right] \quad (2)$$

At deployment time, the policy acts on a new state s_q by retrieving the k -NN of s_q from the training set and aggregating neighbor predictions, $\hat{a} = \hat{f}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_\theta(s_i^*)$. Section 2.4 shows that the parameterization of f is not incidental: it is central to the empirical strength of iMRIL and directly motivates DARP.

This architecture parameterizes the action predictor $\hat{f}$ as an aggregation of learned predictions at training-neighbor states $f(s_i^*)$. Supervising the post-aggregation prediction prevents the learned neighbor predictions from varying arbitrarily, thereby importing the smoothness benefits established in Section 2.2. We make this connection precise by proving the equivalence between iMRIL and Laplacian regularization in Section 2.2.

Equivalence between iMRIL and MRIL: Appendix Section A.1 gives the full equivalence proof between MRIL and iMRIL; the main result and proof sketch appear here.

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). *Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.*

The minimizers of the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) admit the following closed-form expansions

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

iMRIL's neighbor aggregation step applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , preserving low-frequency modes and suppressing high-frequency modes. The alignment between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} establishes iMRIL as a built-in form of Laplacian smoothing (MRIL) with effective $\lambda \approx 1$ in normalized units. Unlike explicit regularization, this implicit filter requires no additional hyperparameter tuning.

Proof sketch. Appendix Section A.1.2 provides the full derivation. The explicit regularizer admits a spectral solution after diagonalizing the k -NN Laplacian, producing the filter $(1 + \lambda \mu)^{-1}$ on each eigenmode. The implicit objective can be written as neighbor aggregation $\hat{f} = S f$ with $S = D^{-1} A$, the random-walk matrix, which shares the same eigenvectors and applies the fixed filter $1 - \mu$. Both filters are low-pass graph filters: small-eigenvalue modes, corresponding to smooth variation across the data manifold, are preserved, while large-eigenvalue modes, corresponding to rapid high-variance neighbor fluctuations, are damped. Therefore iMRIL performs Laplacian smoothing implicitly, reducing variance and enforcing local consistency without tuning λ . $\square$

The implicit Laplacian-smoothing view complements, rather than replaces, policy learning. It constrains the function class represented after aggregation, while the neighbor-conditioned network f_θ learns how expert actions vary under local perturbations and proposes locally adapted actions for each neighbor. Aggregation then smooths these proposals across the neighborhood, reducing variance. Learning supplies accuracy through local bias correction, and aggregation supplies stability through variance control.

2.4 DIFFERENCE-AWARE RETRIEVAL POLICIES: A PRACTICAL INSTANTIATION OF IMRIL FOR IMITATION LEARNING

Building on iMRIL, we now instantiate a practical algorithm for large-scale imitation learning. Starting from Eq. 2, DARP makes two decisive design choices—a difference-based parameterization and a permutation-aware neighborhood aggregation scheme—that drive the empirical performance.

2.4.1 DIFFERENCE-BASED PARAMETERIZATION OF f_θ

Eq. 2 leaves the input representation of f_θ open, and this choice is central. Neighborhood aggregation should learn how expert actions change under local perturbations, so f_θ must observe the *difference* between the query state and each neighbor state. In **Difference-Aware Retrieval Policies (DARP)**, we therefore do not parameterize f_θ only as $f_\theta(s_i^*)$. We also provide the neighbor action a_i^* and the query-neighbor difference $\Delta s_i = s_i^* - s_q$, allowing f_θ to produce a locally adapted candidate action $a'_i = f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)$ for query state s_q and neighbor (s_i^*, a_i^*) .

Let $\mathcal{N}_k(s_q)$ be the index set of the k -nearest neighbors retrieved according to a distance function $d(s_q, s_i^*)$.³ DARP then predicts an action for any query state s_q by applying the neighborhood aggregation principle from Section 2.3 to these difference-aware candidate actions:

$$\hat{a}_q = f_{\text{DARP}}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i \quad (3)$$

$$= \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q). \quad (4)$$

At training time, this architecture yields a direct imitation learning objective on the expert dataset $\mathcal{D}^*$:

$$\arg \min_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\|f_{\text{DARP}}(s_q^*) - a_q^*\|^2] \quad (5)$$

This objective optimizes the predictor parameters f_θ by minimizing the discrepancy between the aggregated action $\hat{a}_q$ and the expert action a_q^* . Because the loss remains simple, f_θ can be implemented with standard feedforward or convolutional networks. Section 3 shows that the difference-based parameterization is crucial for performance. At inference time, DARP executes actions by retrieving k -NN and applying the aggregation operation in Eq. 3.

2.4.2 GOING BEYOND LINEAR AGGREGATION

Simple averaging, $\hat{a}_q = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i$, is only one member of a broader class of permutation-invariant aggregation functions $g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q)})$. More expressive set-compliant models such as Set Transformers (Lee et al., 2019) or DeepSets (Zaheer et al., 2017) can parameterize g_ψ . This gives the generalized prediction rule from Eq. 3, $\hat{a}_q = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$. Beyond expressivity, a parametric aggregation model g_ψ allows DARP to represent richer action distributions, including Gaussian mixture models (Pignat & Calinon, 2019) or diffusion models (Chi et al., 2024), rather than the unimodal Gaussian structure implicit in the L_2 objective of Eq. 5. Instead of predicting $\hat{a}_q$ directly, DARP can predict distribution parameters α for $p(a_q; \alpha)$, such as Gaussian-mixture means, covariances, and weights, or a diffusion score function. This extension enables maximum-likelihood training for multimodal action distributions rather than restricting the policy to unimodal L_2 regression:

$$\arg \max_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\log p(a_q^*; \alpha_\theta(s_q^*))], \text{ where } \alpha_\theta(s_q^*) = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q^*)\}_{i \in \mathcal{N}_k(s_q^*)}) \quad (6)$$

At inference time, DARP samples $\hat{a}_q \sim p(\cdot; \alpha_\theta(s_q))$ after constructing $\alpha_\theta(s_q) = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$ from neighbors retrieved at test time. Appendix Section A.3 gives detailed training pseudocode.

³In our work we use the Euclidean distance in a pre-trained embedding space, although other neighborhood functions are also applicable. We refer the reader to Appendix Section A.2.1 for a thorough discussion of design decisions in constructing neighborhood sets via retrieval.

2.5 SUMMARY

Encouraging local consistency in behavior cloning is manifold regularization, specifically Laplacian smoothing, and it provably improves variance, smoothness, and stability over standard BC (Theorem 1). Rather than inserting an explicit regularizer and tuning its hyperparameter λ , DARP induces the same smoothing effect by building neighborhood aggregation into the policy architecture (Theorem 2). The resulting algorithm is direct: for each expert pair (s_q^*, a_q^*) , DARP retrieves the k -nearest expert states from $\mathcal{D}^*$, computes difference vectors $\Delta s_i = s_i^* - s_q^*$, maps each tuple $(s_i^*, a_i^*, \Delta s_i)$ through f_θ to produce candidate actions, and aggregates those candidates with a permutation-invariant function g_ψ to predict the final action. The method therefore obtains retrieval-based smoothing while training with only the standard imitation learning objective.

3 EXPERIMENTAL EVALUATION

We evaluate DARP around three questions that directly test the paper’s claims: **Q1**: Does DARP consistently outperform standard behavior cloning?, **Q2**: Does DARP remain effective with richer state representations and action distributions?, and **Q3**: Which architectural components drive DARP’s gains? The evaluation spans low-dimensional states, high-dimensional image features, and diverse action representations across continuous control (MuJoCo), robotic manipulation (Robosuite and RoboCasa), and discontinuous environments designed to stress-test neighbor-based policies.

3.1 BASELINE COMPARISONS AND TASK DESCRIPTIONS

MuJoCo Tasks: The MuJoCo (Todorov et al., 2012; Fu et al., 2020) suite tests whether DARP improves continuous-control locomotion. The tasks require controlling different legged embodiments to achieve forward motion on a flat plane: Hopper (single-legged hopping robot), Walker (bipedal humanoid), Ant (quadruped), and HalfCheetah (biped).

Robosuite Tasks: The Robosuite (Zhu et al., 2020) tasks test manipulation policies for a single robotic arm. Stack requires placing a smaller cube on a larger cube, Threading requires inserting a thin needle-like tool into a small ring, and Square Peg requires placing a square wooden block with a center hole onto a square peg.

RoboCasa Tasks: The RoboCasa (Nasiriany et al., 2024) tasks evaluate single-arm manipulation in randomized kitchen scenes. Drawer requires closing an open drawer, Door requires closing an open microwave door, and Stove requires twisting a knob to turn off a burner.

Baseline Comparisons: We compare DARP against baselines that isolate retrieval, parametric learning, and explicit smoothing: (1) **R&P (Sridhar et al., 2025)**: executing the action of the nearest neighbor, (2) **LWR (Pari et al., 2022)**: locally weighted regression over retrieved neighbors, (3) **BC**: standard parametric behavior cloning, (4) **REGENT (Sridhar et al., 2025)**: a transformer-based in-context method conditioned on retrieved neighbors, and (5) **MRIL**: the explicitly smoothed objective from Section 2.2.

3.2 CAN DARP CONSISTENTLY OUTPERFORM STANDARD BEHAVIOR CLONING? (Q1)

We first test the core hypothesis in the cleanest setting: low-dimensional state representations with well-defined and interpretable state distances. The evaluation spans MuJoCo locomotion and Robosuite/RoboCasa manipulation, with manipulation data generated using MimicGen (Mandlekar et al., 2023). In these experiments, the aggregation function g is the simple average of all neighbor action predictions a' .

Method	Hopper	Ant	Walker	HalfCheetah
R&P (Sridhar et al., 2025)	711.82 $\pm$ 85.63	-305.97 $\pm$ 76.42	419.18 $\pm$ 50.21	-178.64 $\pm$ 29.75
LWR (Pari et al., 2022)	1703.78 $\pm$ 245.95	846.59 $\pm$ 216.06	1484.91 $\pm$ 356.54	1945.82 $\pm$ 567.26
BC	2313.65 $\pm$ 203.75	2376.20 $\pm$ 339.43	2658.40 $\pm$ 274.08	1063.23 $\pm$ 371.08
REGENT (Sridhar et al., 2025)	1819.39 $\pm$ 186.24	-302.10 $\pm$ 146.67	507.01 $\pm$ 76.10	169.85 $\pm$ 63.10
MRIL	2793.63 $\pm$ 156.41	3869.08 $\pm$ 241.00	4370.96 $\pm$ 168.13	701.58 $\pm$ 195.08
DARP	3545.57 $\pm$ 3.54	4383.28 $\pm$ 266.37	4894.01 $\pm$ 75.12	5515.41 $\pm$ 841.33
DARP Set Transformer	2965.86 $\pm$ 103.08	4063.79 $\pm$ 218.80	4752.42 $\pm$ 109.23	3417.85 $\pm$ 764.57

Table 1: **DARP and DARP Set Transformer deliver the strongest MuJoCo performance.** Performance comparison of DARP, BC, and retrieval-based baselines on low-dimensional MuJoCo states. Scores are averaged across 100 independent trials with 95% confidence intervals. Parametric policies use Multi-Layer Perceptrons; diffusion-policy results are reported in Section 8.

DARP substantially improves over standard behavior cloning across all tested environments. Robotic manipulation gains range from 15-25 percentage points, and locomotion scores improve markedly as shown in Table 1 and Table 2. Purely non-parametric methods (R&P and LWR) are insufficient on these tasks, and MRIL usually improves over vanilla BC, but the best overall scores in this suite are achieved by the DARP architecture.

Having established the low-dimensional result, we next test whether the practical design from Section 2.4 scales to high-dimensional input representations such as images.

3.3 CAN DARP HANDLE MORE COMPLEX STATE REPRESENTATION AND ACTION DISTRIBUTIONS? (Q2)

High-Dimensional Visual Input Representations. To test DARP beyond compact low-dimensional states, we evaluate simulated robotic manipulation policies using R3M image embeddings (Nair et al., 2022). This setting directly probes whether retrieval remains effective when states are high-dimensional visual feature vectors (Table 3). DARP not only outperforms standard BC,

Method	Stack	Thrd.	Peg
BC	44	38	17
DARP	75	76	52

Table 3: DARP improves success rates (%) on every vision-based Robosuite task.



Figure 3: The “DrawerClose” RoboCasa task, where the robotic arm must close an open drawer.



Figure 4: The “Threading” Robosuite task, where the robotic arm must insert a needle-like implement into a small hole.

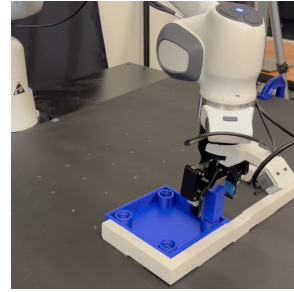


Figure 5: Real-world FurnitureBench square-table assembly task, where the robot must pick up a table leg and screw it into a corner hole of the table-top.

Method	RoboCasa			Robosuite			Real
	Drawer	Door	Stove	Stack	Thrd.	Peg	Sq. Table
BC	54	29	28	47	37	46	44
DARP	85	45	43	72	63	62	92

Table 2: DARP outperforms BC across all three evaluation environments using low-dimensional state features. Scores are success percentages (RoboCasa & Robosuite: 100 trials; Real: 50 trials). DARP wins on every task and more than doubles BC’s real-world score. RoboCasa and Robosuite policies use Multi-Layer Perceptrons, while real-world results use diffusion policy; see Section 8.

but does so by a larger average margin on visual Robosuite tasks ($\sim 35\%$) than on low-dimensional Robosuite tasks ($\sim 22\%$). These results show that DARP adapts more effectively than BC to complex high-dimensional state representations.

Multi-modal Action Distributions. DARP also improves complex multimodal imitation learning: on Push-T, it exceeds behavior cloning by more than 20%. Details appear in Appendix A.2.2.

3.4 HOW DO DIFFERENT ARCHITECTURAL COMPONENTS CONTRIBUTE TO DARP’S PERFORMANCE GAINS? (Q3)

Ablation Study: We isolate the architectural choices responsible for DARP’s gains through a systematic ablation study. The variants are: (1) standard DARP; (2) DARP without neighbor actions; (3) an ensemble of 10 BC agents; (4) DARP with random neighbors instead of distance-based retrieval; (5) DARP using the L_2 norm of the distance vector; (6) the BC baseline using only the query state s_q ; (7) DARP using the query state rather than the query-neighbor distance vector; and (8) DARP with a permutation-dependent, hence **not** permutation-invariant, aggregator over all a ’s. Figure 6 reports the resulting performance.

The ablation study identifies distance vectors and permutation invariance as the main drivers of DARP’s success, while neighbor actions provide a smaller contribution. Random neighbor selection performs poorly, confirming that retrieval must be guided by a meaningful distance metric. The permutation-invariant aggregation function g is also essential: permutation-dependent alternatives sharply degrade performance.

Divergence Analysis: To explain why DARP succeeds where standard BC fails, we analyze rollouts in which BC fails and DARP succeeds. We define the “step of divergence” as the first point where DARP and BC begin receiving significantly different rewards. We define τ_s and τ_Δ as the 1st percentile of training-set likelihoods; equivalently, 1% of training-time deltas are less likely than τ_Δ . Across all six rollouts from two tasks (Robosuite Stack and MuJoCo Hopper), the query state at the

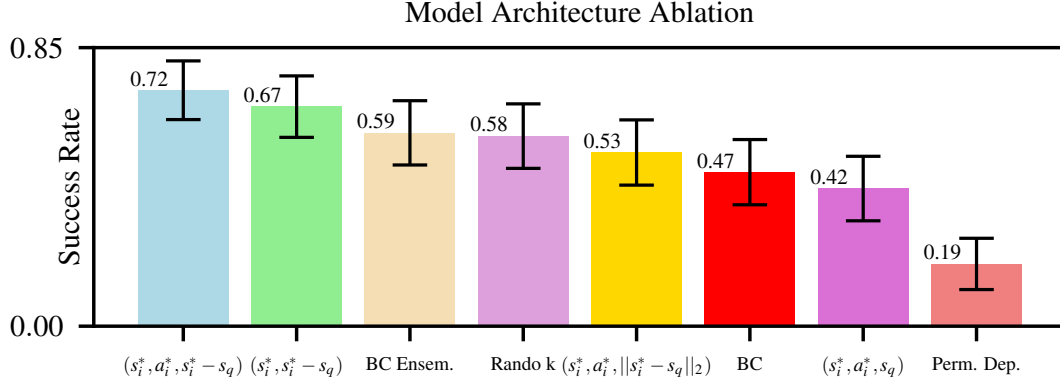


Figure 6: **Distance vectors and permutation invariance are the decisive architectural ingredients in DARP.** Ablating the core architecture shows that the tuple representation $(s_i^*, a_i^*, s_i^* - s_q)$ and permutation-invariant aggregation drive the performance gain. Success rate is averaged across 100 trials on Robosuite Stack with 95% confidence intervals.

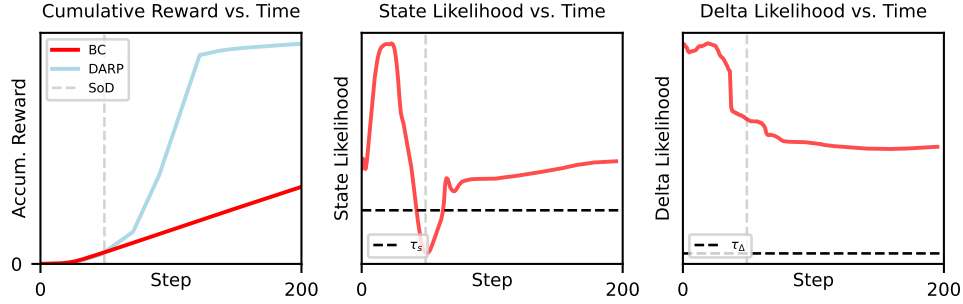


Figure 7: **BC and DARP initially follow the same Robosuite Stack rollout, then diverge exactly where BC fails and DARP succeeds.** The vertical dashed line marks the step of divergence (“SoD”). At the SoD, the state likelihood is $< \tau_s$ (OOD), while the delta likelihood remains $> \tau_\Delta$ (in distribution).

SoD has state likelihood $< \tau_s$ but delta likelihood $> \tau_\Delta$. This pattern directly supports the mechanism behind DARP: the absolute state can be OOD while the neighbor-relative difference remains in distribution, enabling more reliable prediction through difference-aware reparameterization. Figure 7 shows reward drift, SoDs, and state/delta likelihoods for one task; Appendix A.2.4 provides additional robustness experiments.

4 RELATED WORK

Non-Parametric Imitation Learning Methods: Non-parametric IL algorithms demonstrate that local structure is a powerful source of generalization. VINN Pari et al. (2022) uses locally weighted regression for imitation and obtains strong results in image embedding spaces, while MiDiGaP von Hartz et al. (2025) uses mixtures of Gaussian processes to model multimodal trajectories and generalize rapidly. SEABO Lyu et al. (2024) applies retrieval to offline RL by rewarding transitions that remain close to neighbors. FlowRetrieval Lin et al. (2024), STRAP Memmel et al. (2025), and Behavior Retrieval Du et al. (2023) perform non-parametric retrieval and finetuning from large unlabeled datasets, enabling generalization through test-time training. DARP occupies a distinct position: it parameterizes retrieved evidence as $(s_i^*, a_i^*, s_i^* - s_q)$ tuples and learns a semi-parametric policy rather than relying solely on non-parametric aggregation or test-time training. This design directly combines the variance reduction of local methods with the generalization capacity of parametric policies.

Smoothness-Constrained Policy Learning: Recent work has established explicit smoothness constraints as an effective route to policy stability and robustness. L2C2 Kobayashi (2022) studies model-free RL under local Lipschitz continuity constraints, improving smoothness and noise robustness without sacrificing expressiveness, while Asadi et al. (2018) develop related Lipschitz constraints for model-based RL models. CCIL Ke et al. (2024a) extends this principle to imitation learning by generating synthetic corrective labels with a Lipschitz-constrained dynamics model. Related

constraints have also been scaled to humanoid controllers Chen et al. (2024) to reduce deployment shakiness. DARP makes a different and stronger architectural claim: smoothness can be induced implicitly through retrieval-based parameterization while retaining the standard imitation learning objective.

In-Context Learning Methods: Retrieval has also emerged as a natural mechanism for in-context imitation learning. REGENT Sridhar et al. (2025) studies retrieval-augmented generalization by incorporating retrieved states, actions, and rewards into a causal transformer, while DPT Lee et al. (2023) uses supervised transformer pretraining to predict actions from query states and in-context datasets, effectively learning how to explore. Other in-context architectures include ICRT Fu et al. (2024), Instant Policy Vosylius & Johns (2025), and KAT Di Palo & Johns (2024). These methods emphasize rapid adaptation to new tasks and environments; DARP instead targets the standard imitation-learning setting and demonstrates that retrieval can deliver higher performance and stability without changing the supervision regime.

5 CONCLUSION

We introduced **D**ifference-Aware **R**etrieval **P**olicies for Imitation Learning (**DARP**) (DARP), a nearest-neighbor-based algorithm that reparameterizes imitation learning around relative differences between query states and their nearest expert neighbors instead of direct global state-to-action mappings. We prove that this architecture implicitly achieves Laplacian smoothing, giving retrieval a concrete regularization role within standard behavior cloning. Experiments across continuous control and robotic manipulation validate the central claims of the paper. First, DARP consistently outperforms standard behavior cloning with low-dimensional state representations. Second, DARP remains effective across state representations, action-distribution modeling requirements, and task complexities, producing 15-46% improvements across the tested scenarios. Third, architectural ablations identify distance vectors and permutation-invariant aggregation as essential components of the method. Together, these results establish difference-aware retrieval as a practical and principled path to more stable imitation learning without additional data, online feedback, or task-specific supervision.

6 REPRODUCIBILITY STATEMENT

Supplementary source code, policy and environment configuration files, MuJoCo experiment data, scripts for generating expert demonstrations for Robosuite tasks via MimicGen, and modality transformation code will be released upon acceptance to support exact reproduction. On NVIDIA L40 and L40s GPUs, results are expected to match the reported values, except for transformer-based methods (REGENT, Set Transformer), whose non-determinism can produce slight deviations from the numbers in the paper.

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A APPENDIX

A.1 LEMMAS AND PROOFS

A.1.1 PROOF OF THEOREM 1

We prove Theorem 1 by first invoking a standard graph-Laplacian result (Chung, 1997; Zhou et al., 2003; Belkin & Niyogi, 2008).

Lemma 1 (Smoothness regularizer as k -NN graph Laplacian penalty). *Let $\{s_1^*, \dots, s_n^*\}$ be the expert states with corresponding predicted actions $f(s_i^*) \in \mathbb{R}^{d_a}$. For each i , let $\mathcal{N}_k(s_i^*)$ denote the indices of the k -nearest neighbors of s_i^* (excluding i). Define asymmetric weights*

$$\tilde{W}_{ij} = \begin{cases} w_j(s_i^*), & \text{if } j \in \mathcal{N}_k(s_i^*), \\ 0, & \text{otherwise,} \end{cases}$$

and construct a symmetric affinity matrix

$$W_{ij} = \frac{1}{2}(\tilde{W}_{ij} + \tilde{W}_{ji}).$$

Let D be the degree matrix with $D_{ii} = \sum_j W_{ij}$, and define the k -NN graph Laplacian $L = D - W$.

Then the smoothness regularizer can be written as the quadratic form

$$\mathcal{L}_S(f) = \frac{1}{n} \sum_{i=1}^n \sum_{j \in \mathcal{N}_k(s_i^*)} w_j(s_i^*) \|f(s_i^*) - f(s_j^*)\|^2 \propto \text{Tr}(F^\top L F),$$

where $F = [f(s_1^*), f(s_2^*), \dots, f(s_n^*)]^\top \in \mathbb{R}^{n \times d_a}$. Equivalently, in the scalar case,

$$\mathcal{L}_S(f) \propto f^\top L f.$$

Corollary 1 (Continuum limit of smoothness regularizer). *Assume states $\{s_i^*\}_{i=1}^n$ are sampled i.i.d. from a smooth density $p(s^*)$ supported on an m -dimensional C^2 manifold $\mathcal{M} \subset \mathbb{R}^d$. Let W be the symmetrized k -NN affinity matrix constructed from a kernel K_Δ with bandwidth h , and let $L = D - W$ be the graph Laplacian.*

If $n \rightarrow \infty$, $h \rightarrow 0$, and $nh^{m+2} \rightarrow \infty$, then the normalized quadratic form converges to the weighted Dirichlet energy:

$$\frac{1}{n^2 h^{m+2}} \text{Tr}(F^\top L F) \rightarrow C_K \int_{\mathcal{M}} \|\nabla_{\mathcal{M}} f(s)\|_2^2 p(s)^2 d\text{vol}(s),$$

where $C_K > 0$ is a constant depending only on the kernel K_Δ .

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ be the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$. Let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider two estimators trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Then, under the smoothness assumption on f , the following hold:

(i) Variance reduction: *The Laplacian penalty in MRIL acts as a data-dependent Tikhonov regularizer, yielding smaller estimator variance than vanilla BC.*

(ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants with arbitrarily large Lipschitz constants between training states.

(iii) Policy stability: In a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^4$$

accumulates error linearly for vanilla BC, but sublinearly for MRIL, since the smoothness regularizer enforces $\|f(s^*) - f(s'^*)\| = O(\|s^* - s'^*\|)$ for neighbors s^*, s'^* .

Thus MRIL provides stronger generalization and stability guarantees than vanilla BC.

Proof. We establish each claim in turn.

(i) Variance reduction. The Laplacian penalty in $\mathcal{L}_{\text{MRIL}}$ is

$$\sum_{i,j} W_{ij} \|f(s_i^*) - f(s_j^*)\|^2 = 2f^\top Lf,$$

where $L = D - W$ is the graph Laplacian, and W is the k -NN affinity matrix. By Lemma 1, this equals the empirical Dirichlet energy of f on the k -NN graph. It is well known (Zhou et al., 2003; Belkin & Niyogi, 2008) that such a quadratic penalty is equivalent to Tikhonov regularization with respect to the graph Laplacian norm $\|f\|_L^2 = f^\top Lf$. In statistical learning theory, adding a Tikhonov penalty strictly reduces the variance of the estimator compared to the unregularized solution while keeping the bias term controlled. Thus MRIL obtains smaller estimator variance than vanilla BC, which uses no such penalty.

(ii) Smoothness guarantee. Consider the continuum limit (Corollary 1): for i.i.d. samples $\{s_i^*\}$ from density p on a smooth manifold $\mathcal{M}$, the normalized penalty converges to

$$\int_{\mathcal{M}} \|\nabla f(s)\|^2 p(s)^2 d\text{vol}(s).$$

This is the weighted Dirichlet energy of f on $\mathcal{M}$. If this integral is finite, f belongs to the Sobolev space $H^1(\mathcal{M}, p^2)$, and in particular f is locally Lipschitz almost everywhere with

$$\|f(s^*) - f(s'^*)\| \leq C\|s^* - s'^*\| \quad \text{for } p\text{-a.e. neighbor pairs } s^*, s'^*.$$

Therefore minimizers of $\mathcal{L}_{\text{MRIL}}$ have uniformly bounded local Lipschitz constants along high-density regions of the state space. By contrast, minimizers of vanilla BC have no such constraint: any oscillatory interpolant that matches the training data exactly yields the same supervised risk, so arbitrarily large Lipschitz constants are possible.

(iii) Policy stability. Let $\Delta_t = \|s_t - s_t^*\|$ denote the deviation at time t . For Lipschitz dynamics T ,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|.$$

Decompose the action error:

$$\|f(s_t) - f^*(s_t^*)\| \leq \|f(s_t) - f^*(s_t)\| + \|f^*(s_t) - f^*(s_t^*)\|.$$

For vanilla BC, the first term $\|f(s_t) - f^*(s_t)\|$ is only minimized on the empirical distribution $P_{\mathcal{S}}$; off-distribution, it may be $O(1)$ regardless of Δ_t . The second term satisfies $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Hence the recursion can take the form

$$\Delta_{t+1} \leq L_s \Delta_t + L_a (O(1) + O(\Delta_t)),$$

which accumulates linearly in t .

For MRIL, the Laplacian penalty enforces

$$\|f(s^*) - f(s'^*)\| \leq C\|s^* - s'^*\| \quad \text{for neighbor pairs } (s^*, s'^*),$$

⁴ L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

as shown in part (ii). Thus $\|f(s_t) - f^*(s_t)\| = O(h^2)$ (where h is the radius of the kernel bandwidth around the point s_t) by local-linear regression error bounds, and $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Combining these,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a (O(h^2) + O(\Delta_t)).$$

Since the constant multiplying Δ_t is strictly smaller under the smoothness constraint, the cumulative error grows strictly slower than in vanilla BC. In particular, error growth is sublinear in the rollout horizon when h is small, whereas it is linear for vanilla BC.

Conclusion. Claims (i)–(iii) establish that MRIL yields lower variance, uniform smoothness control, and sublinear rollout error accumulation compared to vanilla BC, completing the proof. $\square$

Kernel choice. For the IC smoothness regularizer, we adopt a Gaussian kernel

$$w_i(s^*) \propto \exp\left(-\frac{\|s^* - s_i^*\|^2}{2h^2}\right), \quad \sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) = 1,$$

with bandwidth h set to the median distance to the k -th nearest neighbor across the dataset. This choice is standard in manifold regularization (Belkin & Niyogi, 2008; Zhou et al., 2003) and ensures that the graph Laplacian penalty converges to the Dirichlet energy in the continuum limit. In practice, this default remains stable across tasks, though other kernels (e.g., uniform k -NN or exponential decay) yield qualitatively similar results.

A.1.2 PROOF OF THEOREM 2

We begin with lemmas that establish the spectral form of explicit Laplacian regularization and neighbor aggregation.

Lemma 2 (Spectral form of explicit Laplacian regularization). *Let L be the symmetric normalized graph Laplacian with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$. The minimizer of the penalized objective*

$$\mathcal{L}_\lambda(f) = \|f - a^*\|^2 + \lambda f^\top L f$$

has the closed-form expansion

$$f_\lambda = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j.$$

Thus λ directly controls the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$ applied to each Laplacian mode.

Proof. Diagonalize $L = U \Lambda U^\top$ with $U = [u_1, \dots, u_n]$ orthogonal and $\Lambda = \text{diag}(\mu_1, \dots, \mu_n)$. Write $f = U c$, $a^* = U b$ in this basis. The objective becomes

$$\|U c - U b\|^2 + \lambda c^\top \Lambda c = \sum_{j=1}^n (c_j - b_j)^2 + \lambda \mu_j c_j^2.$$

Minimizing each term yields $c_j = \frac{1}{1 + \lambda \mu_j} b_j$. Transforming back gives the stated expansion. $\square$

Lemma 3 (Spectral form of neighbor aggregation). *Let $S = D^{-1}A$ be the random-walk matrix of the k -NN graph, with adjacency A and degree D . For any prediction vector f , the neighbor-averaged prediction is $\hat{f} = S f$. In the Laplacian eigenbasis, this corresponds to the spectral filter*

$$\hat{f} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j,$$

i.e. $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

Proof. By definition, $L = I - D^{-1/2} A D^{-1/2}$ and $S = D^{-1} A = I - L_{\text{rw}}$ where $L_{\text{rw}} = D^{-1} L$ is the random-walk Laplacian. Since L_{rw} and L share the same spectrum up to similarity transform, the eigenbasis $\{u_j\}$ diagonalizes S . Thus for each mode u_j , $S u_j = (1 - \mu_j) u_j$, yielding the claimed spectral filter. $\square$

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.

The minimizers of the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) admit the following closed-form expansions

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

iMRIL's neighbor aggregation step applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , preserving low-frequency modes and suppressing high-frequency modes. The alignment between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} establishes iMRIL as a built-in form of Laplacian smoothing (MRIL) with effective $\lambda \approx 1$ in normalized units. Unlike explicit regularization, this implicit filter requires no additional hyperparameter tuning.

Proof. From Lemma 3, the neighbor aggregation operator $S = D^{-1}A$ acts on Laplacian eigenmodes u_j as

$$Su_j = (1 - \mu_j)u_j,$$

where μ_j are the normalized Laplacian eigenvalues. Thus, in the graph Fourier basis, neighbor aggregation implements multiplying each mode by the fixed spectral filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

On the other hand, Lemma 2 shows that explicit Laplacian regularization with parameter λ yields the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$. Both filters downweight high-frequency modes ($\mu \gg 0$) while preserving low-frequency modes ($\mu \approx 0$). The key difference is that $\phi_\lambda(\mu)$ requires tuning λ , whereas $\phi_{\text{DARP}}(\mu)$ is parameter-free.

To see the equivalence, note that for small μ ,

$$\phi_{\text{DARP}}(\mu) = 1 - \mu \approx (1 + \mu)^{-1} = \phi_{\lambda=1}(\mu) \quad \text{up to } O(\mu^2) \text{ terms.}$$

Thus DARP can be interpreted as performing Laplacian smoothing with an effective regularization weight of order $\lambda \approx 1$ in normalized units. For large μ , $\phi_{\text{DARP}}(\mu)$ damps high-frequency modes even more strongly by driving them toward zero, giving a sharper low-pass effect than explicit regularization.

Therefore, DARP's aggregation step is mathematically equivalent to implicit Laplacian regularization with the fixed spectral filter ϕ_{DARP} , eliminating the need to tune λ explicitly. $\square$

DARP is therefore a form of *locally adaptive implicit regularization*: instead of introducing a global weight λ , neighbor aggregation enforces smoothness through the graph structure itself. The effective

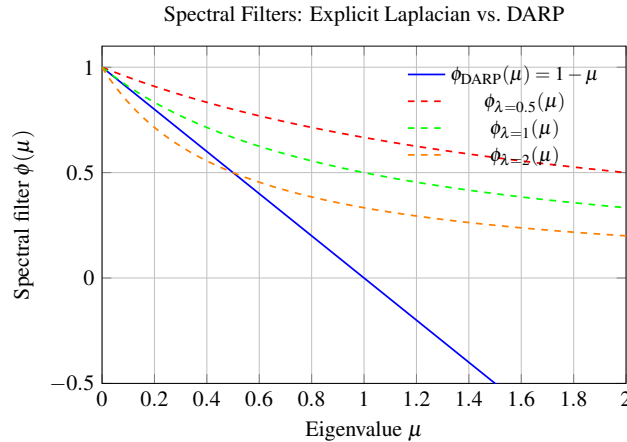


Figure 8: DARP achieves sharper low-pass filtering without tuning an explicit regularization weight.

regularization strength varies with local degree and neighborhood geometry, adapting to the density of expert demonstrations. Spectrally, this is the fixed filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$, which suppresses high-frequency modes more aggressively than any fixed explicit λ . Figure 8 illustrates how DARP achieves sharper low-pass filtering without hyperparameter tuning.

A.2 ADDITIONAL EXPERIMENTAL DETAILS

A.2.1 RETRIEVAL

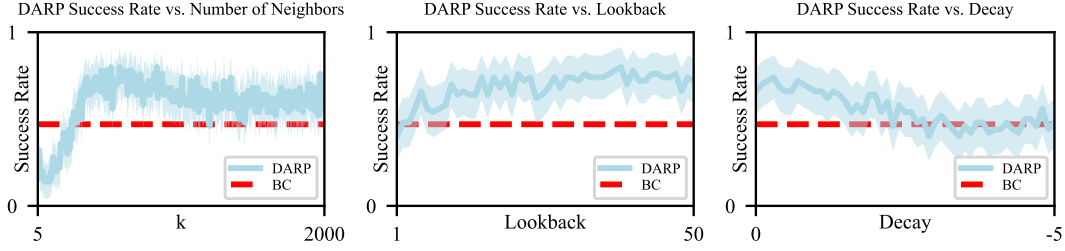


Figure 9: Retrieval hyperparameter sweep for DARP: (left) performance is weak with too few neighbors, reaches a global optimum around 500 neighbors, and plateaus slightly above BC as k approaches the dataset size; (center) performance generally improves as more history is included and falls below BC only with little or no history; (right) performance depends on how strongly older observations are decayed during retrieval. When decay is too high, DARP behaves similarly to little or no lookback and falls below BC. Success rate is measured over 50 trials on Robosuite Stack with 95% confidence intervals.

The distance function $d(s_q, s_i^*)$ is a central retrieval design choice. Simple Euclidean distance between states can work, but our experiments use a sequence-aware variant that incorporates recent history into the distance calculation.

Let the query trajectory be $S_q = (\dots, s_{q,-1}, s_{q,0})$, where $s_{q,0}$ is the current query state s_q . To compute $d(s_q, s_i^*)$ for an expert state s_i^* , we identify its source trajectory S_j^* and its index t , rewriting it as $s_{j,t}^*$. With a lookback parameter ℓ that specifies how many past states are included, the distance is:

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\|$$

This distance accumulates Euclidean distances between the current and previous $\ell - 1$ states in the query and source trajectories, assuming valid indices. In practice, recent states should influence retrieval more strongly, so we apply exponential decay with rate $\gamma \geq 0$:

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\| \cdot e^{-\gamma n}$$

Figure 9 reports how success changes as these retrieval parameters are swept.

A.2.2 CAN DARP HANDLE TASKS REQUIRING THE REPRESENTATION OF MULTI-MODAL ACTION DISTRIBUTIONS?

We test whether DARP handles complex action distributions on Push-T (Chi et al., 2024), which requires multimodal action modeling (Figure 10). In this experiment, DARP uses a Set Transformer head to predict Gaussian Mixture Model parameters. With this GMM head, DARP handles multimodality effectively and improves over BC by 22% on Push-T (Q2) (Table 4). This confirms that DARP extends naturally to multimodal action-distribution modeling.

A.2.3 CAN DARP HANDLE DISCONTINUOUS ENVIRONMENTS WHERE NEARBY STATES MAY REQUIRE OPPOSING ACTIONS?

A key concern for neighbor-based approaches is whether they remain reliable in discontinuous environments, where Euclidean-near states may require sharply different actions. We address this concern with a stress test based on a modified D4RL Umaze environment (Figure 11).

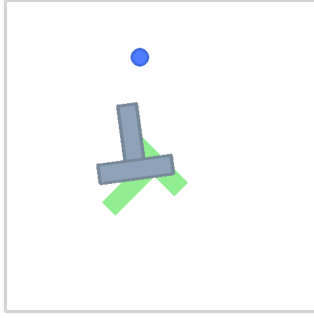


Figure 10: **Push-T Environment.** The policy controls the blue circle to push the T-shaped block.

Method	Score
BC	48 ± 8
DARP	70 ± 8

Table 4: **Push-T Results.** Averaged over 100 trials, DARP clearly outperforms BC.

Even in this deliberately challenging discontinuous environment, DARP achieves a 57% success rate compared to BC’s 25%. (Q3) (see Table 5) This result shows that distance vectors and permutation-invariant aggregation help the model distinguish useful from misleading neighbors, even when spatial proximity does not guarantee action similarity.

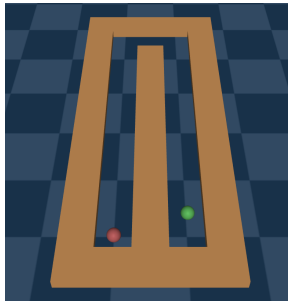


Figure 11: **Long maze environment.** The policy must move a force-actuated ball from the green start to the red destination.

Method	Succ. (%)
BC	25
DARP	57

Table 5: **Long maze results.** Averaged over 100 trials, DARP substantially outperforms BC.

A.2.4 CAN DARP RECOVER FROM BC ERROR?

To analyze robustness to accumulated error, we roll out a BC agent in an environment where it fails, fork the environment every k steps, and start a DARP rollout from each clone. Fig. 12 shows that even as BC approaches failure and drifts away from expert-demonstration support, DARP often recovers and achieves high scores. This provides direct evidence that DARP is more robust to accumulated error and remains effective in the slightly out-of-distribution states reached by failing BC rollouts.

A.2.5 COMPARISON WITH CCIL

Table 6 compares DARP with CCIL, a baseline that explicitly induces smoothness. Using the reported CCIL scores, we compare percent improvement over BC. DARP outperforms CCIL on three of four environments and matches it on the fourth, with a particularly large HalfCheetah margin (418.7% vs 5.4% improvement).

A.2.6 DISTANCE METRIC SENSITIVITY

Although the main experiments use Euclidean distance for neighbor retrieval, high-dimensional embeddings such as R3M naturally invite alternatives such as cosine similarity. Table 7 shows that

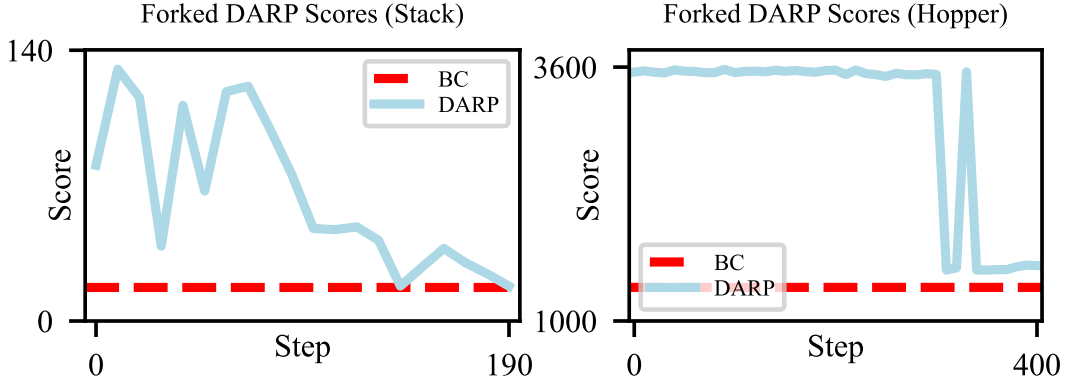


Figure 12: In Robosuite Stack and MuJoCo Hopper, we roll out a failing BC agent and fork the environment every k steps (here, $k = 10$). Even near the end of the failing BC rollout, DARP recovers to high scores; recovery is prevented only about halfway through Stack and about 80% through Hopper.

Environment	Method	Improvement over BC (%)
Hopper	CCIL	32.6%
	DARP	53.2%
Walker	CCIL	84.1%
	DARP	84.1%
Ant	CCIL	12.6%
	DARP	84.5%
HalfCheetah	CCIL	5.4%
	DARP	418.7%

Table 6: **Relative improvement over BC compared to CCIL.** On standard MuJoCo benchmarks, DARP achieves higher or equal relative gains across all tasks.

Method	Success Rate (%)
BC	47
DARP (Cosine Similarity)	70
DARP (Euclidean Distance)	75

Table 7: **Distance Metric Sensitivity.** With R3M features on Robosuite Stack, DARP remains strong under both Euclidean distance and cosine similarity, beating BC by 28% and 23% respectively.

DARP remains strong with cosine similarity, only 5% below Euclidean distance. This supports the robustness of DARP to reasonable distance-metric choices.

A.2.7 DARP IN COMBINATION WITH DIFFUSION POLICY

Although the main experiments use an MLP backbone, diffusion policy (Chi et al., 2024) is a strong imitation-learning model class, especially for manipulation. Table 8 shows that DARP is complementary to diffusion and can be combined with it for stronger performance. With a diffusion backbone, DARP improves over the MLP-backed DARP model by 20%, 34%, and 26% success-rate points across the three tasks, and improves over standard BC by 48%, 60%, and 38%.

Table 9 confirms that DARP with an MLP backbone is much faster than standard diffusion, especially at inference time.

Method	Stack	Threading	Peg Insertion
BC (MLP)	34	12	28
DARP (MLP)	62	38	40
Diffusion	52	44	54
DARP w/ Diffusion	82	72	66

Table 8: **DARP in combination with diffusion policy.** DARP improves both standard MLP policies and diffusion policies. Results report success rates across 50 trials on state-based Robosuite Stack, Threading, and Peg Insertion tasks. The number of demonstrations is much smaller than in Table 2, so baseline BC and DARP numbers do not match.

Method	Training (s/epoch)	Inference (s/step)
Diffusion	5.610	0.15700
DARP (MLP)	0.559	0.00437

Table 9: **Computational efficiency of DARP with an MLP backbone and diffusion policy.** Runtime comparison between DARP (MLP) and diffusion policy. Diffusion policy is substantially more expensive, running $\approx 10\times$ slower during training and $\approx 36\times$ slower during inference.

Method	Success Rate (%)
BC	45
DARP	60

Table 10: **Results with human demonstrations.** DARP preserves its advantage when trained on human data rather than demonstrations collected by RL policies. Results are on Robosuite Stack.

A.2.8 DARP TRAINED ON HUMAN DEMONSTRATIONS

The expert demonstrations used for MuJoCo and Robosuite training are collected by an optimal Reinforcement Learning policy. To verify that the gain is not tied to RL-generated demonstrations, we also train on human demonstrations. On Robosuite Stack, DARP outperforms standard BC by 15% with human data (Table 10), confirming that the advantage transfers to human demonstration settings.

A.2.9 CHOICE OF RETRIEVAL HYPERPARAMETERS

Environment	Method	Score (Mean)
Hopper	BC	507.49
	DARP	805.69
Stack	BC	0.12
	DARP	0.31

Table 11: **Performance comparison when validation loss selects training epochs and retrieval hyperparameters.** Mean scores on Hopper and Stack show that DARP maintains a clear advantage over BC.

Figure 9 shows that poor retrieval settings, such as using very few neighbors, can make DARP underperform standard BC. Validation loss nevertheless provides an effective selection rule for performant retrieval hyperparameters, and Table 11 shows that this rule preserves the DARP advantage.

A.3 PSEUDOCODE

Algorithm 1 provides the full DARP training pseudocode.

Algorithm 1 Difference-Aware Retrieval Policies

```

1: Input: Expert demonstrations  $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$ , number of neighbors  $k$ 
2: Initialize:  $f$  parameters  $\theta$ 
3: if  $g$  is parametric then
4:   Initialize:  $g$  parameters  $\psi$ 
5: end if
6: // Training Loop
7: while not converged do
8:   Sample batch of query data  $(s_q^*, a_q^*) \sim \mathcal{D}^*$ 
9:   for each query pair  $(s_q^*, a_q^*)$  in batch do
10:    // Find  $k$ -Nearest Neighbors from the entire dataset  $\mathcal{D}^*$ 
11:     $\mathcal{N}_k(s_q^*) \leftarrow \text{argmin-}k; d(s_q^*, s_j^*)$ 
12:    // Compute Neighbor-based Predictions
13:    for each neighbor index  $i \in \mathcal{N}_k(s_q^*)$  do
14:       $a'_i \leftarrow f_\theta(s_i^*, a_i^*, s_i^* - s_q^*)$ 
15:    end for
16:    // Aggregate Predictions
17:    if  $g$  is parametric then
18:       $\hat{a}_q \leftarrow g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
19:    else
20:       $\hat{a}_q \leftarrow g(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
21:    end if
22:  end for
23:  // Update Parameters based on the batch loss
24:   $\mathcal{L} \leftarrow \sum_{(s_q^*, a_q^*) \in \text{batch}} \|\hat{a}_q - a_q^*\|^2$ 
25:  // Gradient descent step
26:   $\theta \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}$ 
27:  if  $g$  is parametric then
28:     $\psi \leftarrow \psi - \alpha \nabla_\psi \mathcal{L}$ 
29:  end if
30: end while
31: Output: Trained parameters  $\theta$  and, if applicable,  $\psi$ 

```

A.4 RUNTIME ANALYSIS

Environment	k	Train (s/epoch)	Test (s/step)
Hopper (1,000 datapoints)	BC	0.102	0.00127
	100	0.126	0.00413
	200	0.128	0.00418
	300	0.130	0.00420
	400	0.131	0.00421
	500	0.130	0.00419
Stack (4,200 datapoints)	BC	0.431	0.00139
	100	0.556	0.00437
	250	0.539	0.00442
	500	0.559	0.00437
	750	0.576	0.00433
	1000	0.591	0.00478
	1500	0.682	0.00452
	2000	0.758	0.00451

Table 12: **Runtime comparison across environments.** Training and testing speeds (in seconds) for Behavior Cloning (BC) and varying values of k on Hopper and Stack datasets show that DARP remains practical as retrieval size increases.

Table 12 quantifies the training-time and inference-time overhead of DARP. Computation scales sub-linearly as k increases, and inference remains fast enough for real-time robotic control. For example, on Robosuite Stack with $k = 500$, inference takes approximately 0.00437 seconds per step on our hardware, corresponding to a control frequency above 230 Hz.